

AICE2010 Tutorial Sheet 2

March 6, 2026

Question 1

In terms of the cut-off frequency ω_0 obtain the transfer function of a fourth order low-pass Butterworth filter

Question 2

It is required to construct a low-pass Butterworth filter having a cut-off frequency of 1 kHz. Determine the pole locations and hence the transfer functions for both a third order and a fourth order filter. Write each transfer function for each system. Compare the attenuation of the two filters (in dB) at a frequency equal to twice the cut off frequency.

Question 3

A low pass Butterworth filter is to have a cut off frequency of 20 Hz. Its attenuation at three times the cut off frequency is to be at least 25 dB. Obtain the transfer function of the lowest order filter that will meet this specification.

Question 4

In terms of the cut off frequency ω_0 , obtain the transfer function of a fourth order high pass Butterworth filter.

Question 5

Obtain the transfer function of a third order high pass filter with cut off frequency at 200 Hz. Evaluate the magnitude of the filter at frequencies equal to twice and half the cut off frequency respectively.

Question 6

A particular application requires a high-pass analogue Butterworth filter of order 3 with pass-band edge at 1 kHz. Design this filter and leave your answer in the form of the required transfer function $G(s)$, i.e., as the ratio of real polynomials in s . Also compute the magnitude response of the filter in dB at half the pass-band edge frequency. Is this value likely to be acceptable in many applications?

Question 7

Consider the transfer-functions

$$G_1(s) = \frac{ks}{s+a}, \quad G_2(s) = \frac{kb}{s+b}$$

where a and b are positive real scalars where $b > a$. Consider the transfer function

$$G(s) = G_1(s) + G_2(s)$$

Justify this claim by approximate Bode gain plots that $G(s)$ is the transfer function of a band pass filter. What choices of $G_1(s)$ and $G_2(s)$ would result in $G(s)$ being the transfer-function of a band stop filter?

Question 8

Design a third order high-pass Butterworth filter with cut-off frequency of 400 Hz. Give your answer in transfer-function form. Evaluate the magnitude of the frequency response of the filter at five times the cutoff frequency and at half the cut-off frequency. Also design a bandpass filter with pass-band frequencies a and b Hz where $a < b$.